

mod N 同余类 (2)

已经证明了: $(\mathbb{Z}/N\mathbb{Z}, +, \cdot, [0], [1])$ 是交换环.

Lemma: 对 $\forall N \in \mathbb{Z}_{\geq 1}$, 有: $(\mathbb{Z}/N\mathbb{Z})^{\times} = \{[x] : x \in \mathbb{Z} \text{ 且 } \gcd(x, N) = 1\}$

Proof: 之前已经证明了 $\mathbb{Z}/N\mathbb{Z} = \{[x] : x \in \mathbb{Z}\} = \{[0], [1], \dots, [N-1]\}$

记 $A = \{[x] : x \in \mathbb{Z} \text{ 且 } \gcd(x, N) = 1\}$. 当 $N \in \mathbb{Z}_{\geq 2}$ 时, 有:

对 $\forall [x] \in (\mathbb{Z}/N\mathbb{Z})^{\times}$ (其中 $x \in \mathbb{Z}$), $\exists [y] \in (\mathbb{Z}/N\mathbb{Z})^{\times}$ (其中 $y \in \mathbb{Z}$),

使得: $[x] \cdot [y] = [1] = [y] \cdot [x]$.

$$\therefore [xy] = [1] = [yx]$$

$$\therefore yx \equiv 1 \pmod{N} \quad \therefore \gcd(x, N) = 1 \quad \therefore [x] \in A$$

$$\therefore (\mathbb{Z}/N\mathbb{Z})^{\times} \subseteq A$$

对 $\forall [x] \in A$ (其中 $x \in \mathbb{Z}$ 且 $\gcd(x, N) = 1$) $\therefore \gcd(x, N) = 1$

$$\therefore \exists c \in \mathbb{Z}, \text{ s.t. } cx \equiv 1 \pmod{N} \quad \therefore [cx] = [1]$$

$$\therefore c \in \mathbb{Z} \quad \therefore [c] \in \mathbb{Z}/N\mathbb{Z}$$

$$\therefore [c][x] = [cx] = [1], \quad [x][c] = [xc] = [cx] = [1]$$

整数乘法的交换律

$$\therefore [c][x] = [1] = [x][c]$$

$$\therefore [x] \in (\mathbb{Z}/N\mathbb{Z})^{\times} \quad \therefore A \subseteq (\mathbb{Z}/N\mathbb{Z})^{\times} \quad \therefore (\mathbb{Z}/N\mathbb{Z})^{\times} = A$$

$$\text{当 } N=1 \text{ 时, } \mathbb{Z}/N\mathbb{Z} = \mathbb{Z}/1\mathbb{Z} = \{\mathbb{Z}\} \quad \therefore (\mathbb{Z}/N\mathbb{Z})^{\times} = \{\mathbb{Z}\}$$

$$\therefore \text{对 } \forall x \in \mathbb{Z}, \text{ 有: } [x]_1 = \mathbb{Z} \quad \therefore A = \{\mathbb{Z}\} \quad \therefore (\mathbb{Z}/N\mathbb{Z})^{\times} = A. \quad \square$$

Lemma: 对 $\forall N \in \mathbb{Z}_{\geq 1}$, 有: $|(\mathbb{Z}/N\mathbb{Z})^\times| = \varphi(N)$

Proof: 由上引理可得:

$$(\mathbb{Z}/N\mathbb{Z})^\times = \{[x] : x \in \mathbb{Z} \text{ 且 } \gcd(x, N) = 1\}$$

$$\stackrel{\textcircled{1}}{=} \{[x] : x \in \mathbb{Z} \text{ 且 } 1 \leq x \leq N \text{ 且 } \gcd(x, N) = 1\}$$

关于等号①的论证:

$$\text{令 } A = \{[x] : x \in \mathbb{Z} \text{ 且 } \gcd(x, N) = 1\}, B = \{[x] : x \in \mathbb{Z} \text{ 且 } 1 \leq x \leq N \text{ 且 } \gcd(x, N) = 1\}$$

$$\text{对 } \forall [x] \in B \text{ (其中 } x \in \mathbb{Z} \text{ 且 } 1 \leq x \leq N \text{ 且 } \gcd(x, N) = 1), \because x \in \mathbb{Z} \text{ 且 } \gcd(x, N) = 1$$

$$\therefore [x] \in A \quad \therefore B \subseteq A.$$

对 $\forall [x] \in A$ (其中 $x \in \mathbb{Z}$ 且 $\gcd(x, N) = 1$), 有:

$$\because x \in \mathbb{Z} \quad \therefore \exists \lambda \in \mathbb{Z} \text{ 且 } 1 \leq \lambda \leq N, \text{ s.t. } x \equiv \lambda \pmod{N} \quad \therefore [x] = [\lambda]$$

$$\because \gcd(\lambda, N) = \gcd(x, N) = 1 \quad \therefore [\lambda] = [\lambda] \in B \quad \therefore A \subseteq B \quad \therefore A = B.$$

$\therefore 1, 2, \dots, N-1, N$ 关于 $\text{mod } N$ 两两不同余

$\therefore [1], [2], \dots, [N-1], [N]$ 是彼此不同的元素.

$$\therefore |(\mathbb{Z}/N\mathbb{Z})^\times| = |\{[x] : x \in \mathbb{Z} \text{ 且 } 1 \leq x \leq N \text{ 且 } \gcd(x, N) = 1\}|$$

$$= \varphi(N)$$



Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有:

$$\varphi(n) = n-1 \iff n \text{ 是素数}$$

Proof: ($\Leftarrow$): $\because n$ 是素数 $\therefore \varphi(n) = n-1$

($\Rightarrow$): 假设 $n=1$, 则有: $\varphi(n) = \varphi(1) = 1$.

$$\therefore \varphi(n) = n-1 = 1-1 = 0 \quad \therefore 1 = \varphi(n) = 0 \text{ 矛盾.}$$

$$\therefore n \neq 1 \quad \therefore n \in \mathbb{Z}_{\geq 2} \quad \therefore \gcd(n, n) = n \geq 2 > 1$$

$\therefore \varphi(n) = n-1$ $\therefore$ 不超过 n 而与 n 互素的正整数有 $n-1$ 个

$\therefore n$ 与 n 不互素 $\therefore 1, 2, \dots, n-1$ 都与 n 互素.

$\therefore 2 \nmid n$ 且 $3 \nmid n$ 且 $\dots$ 且 $n-1 \nmid n \quad \therefore n$ 是素数 $\square$

Lemma: 对 $\forall N \in \mathbb{Z}_{\geq 1}$, 有:

$$\mathbb{Z}/N\mathbb{Z} \text{ 是域} \iff \varphi(N) = N-1 \iff N \text{ 是素数}$$

Proof: 中间 $\iff$ 右边 上一引理已证.

~~推~~ $\mathbb{Z}/N\mathbb{Z}$ 是域 $\iff \mathbb{Z}/N\mathbb{Z}$ 是交换除环 $\iff \mathbb{Z}/N\mathbb{Z}$ 是除环

$$\iff (\mathbb{Z}/N\mathbb{Z})^{\times} = (\mathbb{Z}/N\mathbb{Z}) \setminus \{[0]\}$$

$$\stackrel{\textcircled{1}}{\iff} |(\mathbb{Z}/N\mathbb{Z})^{\times}| = N-1$$

$$\iff \varphi(N) = N-1 \quad \therefore \text{左边} \iff \text{中间.}$$

关于①的论证:

$$\textcircled{1}(\Rightarrow): \because (\mathbb{Z}/N\mathbb{Z})^\times = (\mathbb{Z}/N\mathbb{Z}) \setminus \{[0]\} = \{[1], [2], \dots, [N-1]\}$$

$$\therefore |(\mathbb{Z}/N\mathbb{Z})^\times| = N-1$$

$$\textcircled{1}(\Leftarrow): \because (\mathbb{Z}/N\mathbb{Z})^\times \subseteq \mathbb{Z}/N\mathbb{Z} = \{[0], [1], \dots, [N-1]\}$$

$$\text{又} \because |(\mathbb{Z}/N\mathbb{Z})^\times| = N-1$$

$\therefore$ 假设 $[0] \in (\mathbb{Z}/N\mathbb{Z})^\times$, 则有 $\mathbb{Z}/N\mathbb{Z} = \{[0]\}$

$$\therefore (\mathbb{Z}/N\mathbb{Z})^\times = \{[0]\} \quad \therefore N=2 \quad \therefore \mathbb{Z}/N\mathbb{Z} \text{ 不是零环. 矛盾.}$$

$$\therefore [0] \notin (\mathbb{Z}/N\mathbb{Z})^\times$$

$$\therefore (\mathbb{Z}/N\mathbb{Z})^\times = \{[1], [2], \dots, [N-1]\} = (\mathbb{Z}/N\mathbb{Z}) \setminus \{[0]\} \quad \square$$